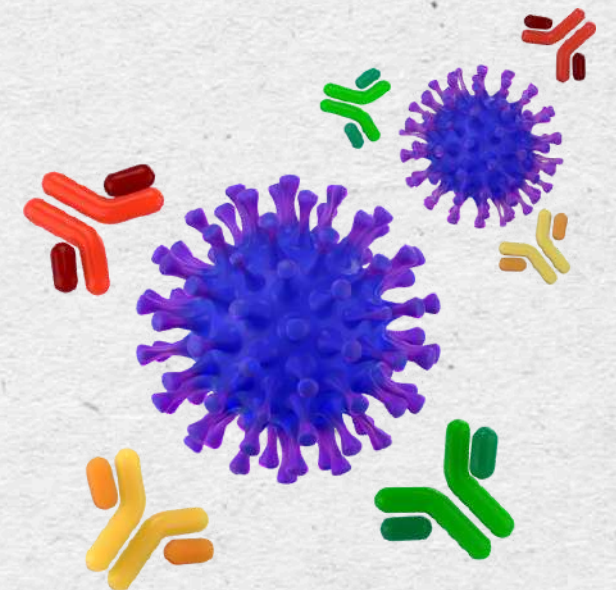


LAB 4

ANTIBODY PURIFICATION BY AMMONIUM SULFATE PRECIPITATION

IMMUNOLOGY & SEROLOGY LABORATORY
(BT336)



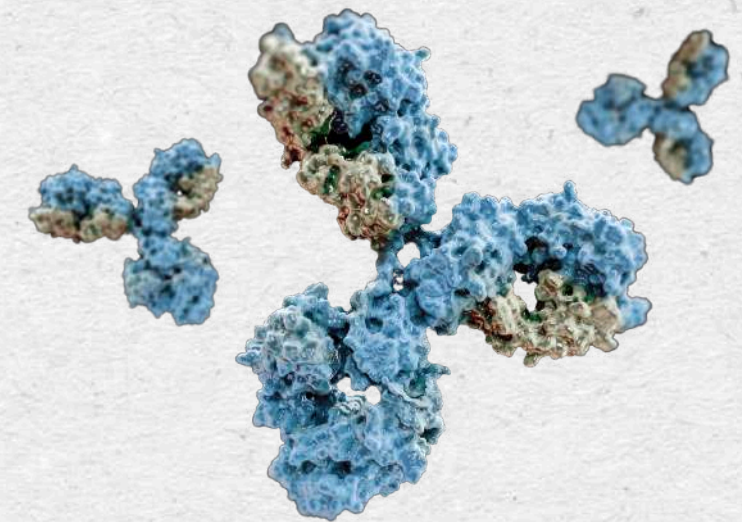
INTRODUCTION



- **Antibodies**, also known as **immunoglobulins**, are specialized proteins produced by the immune system in response to the presence of foreign substances called antigens.
 - They play crucial roles in defending the body against pathogens such as bacteria, viruses, parasites, and toxins.
 - **Antibodies** are highly specific and capable of recognizing and binding to specific antigens with remarkable precision.
-



INTRODUCTION



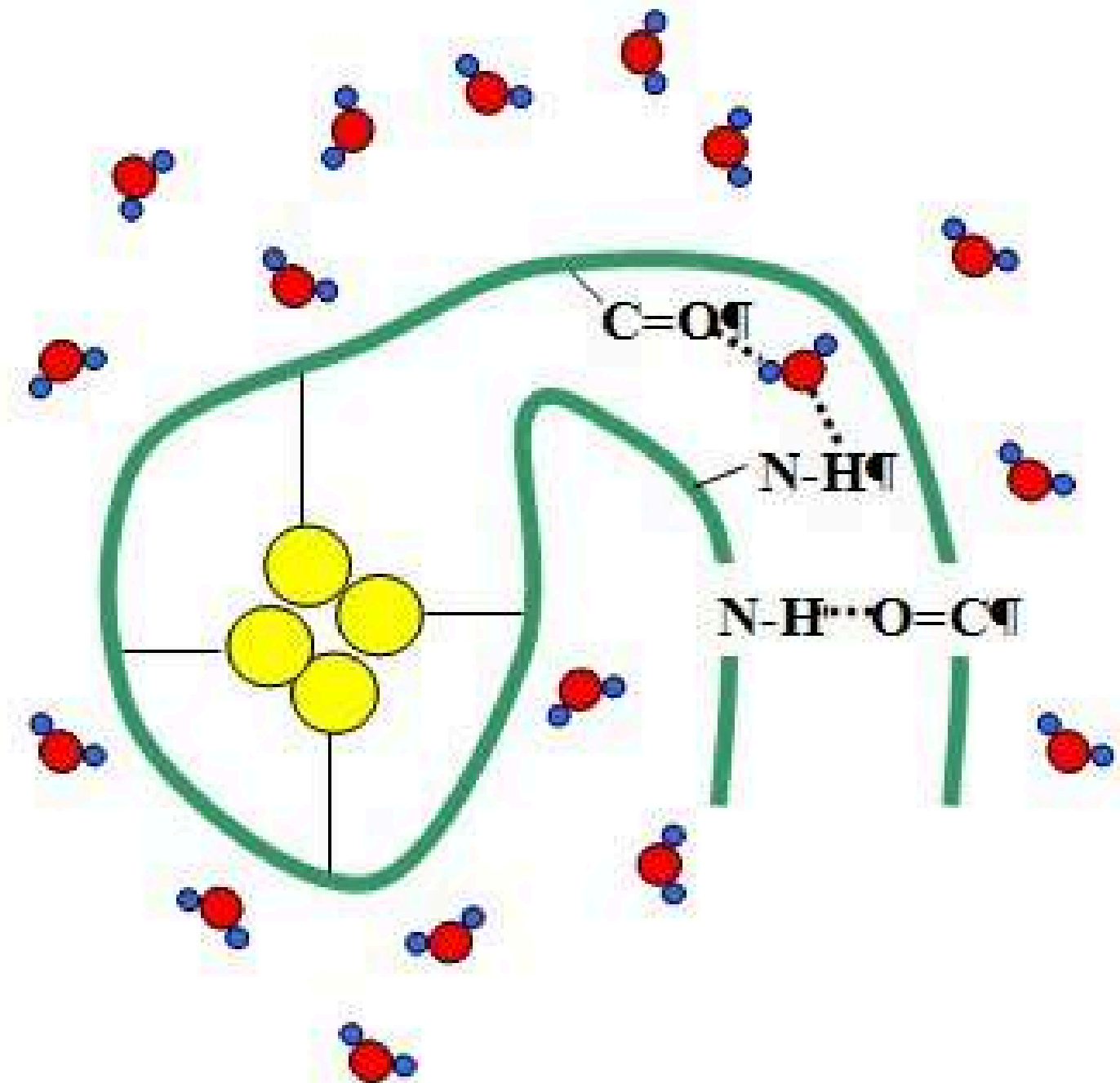
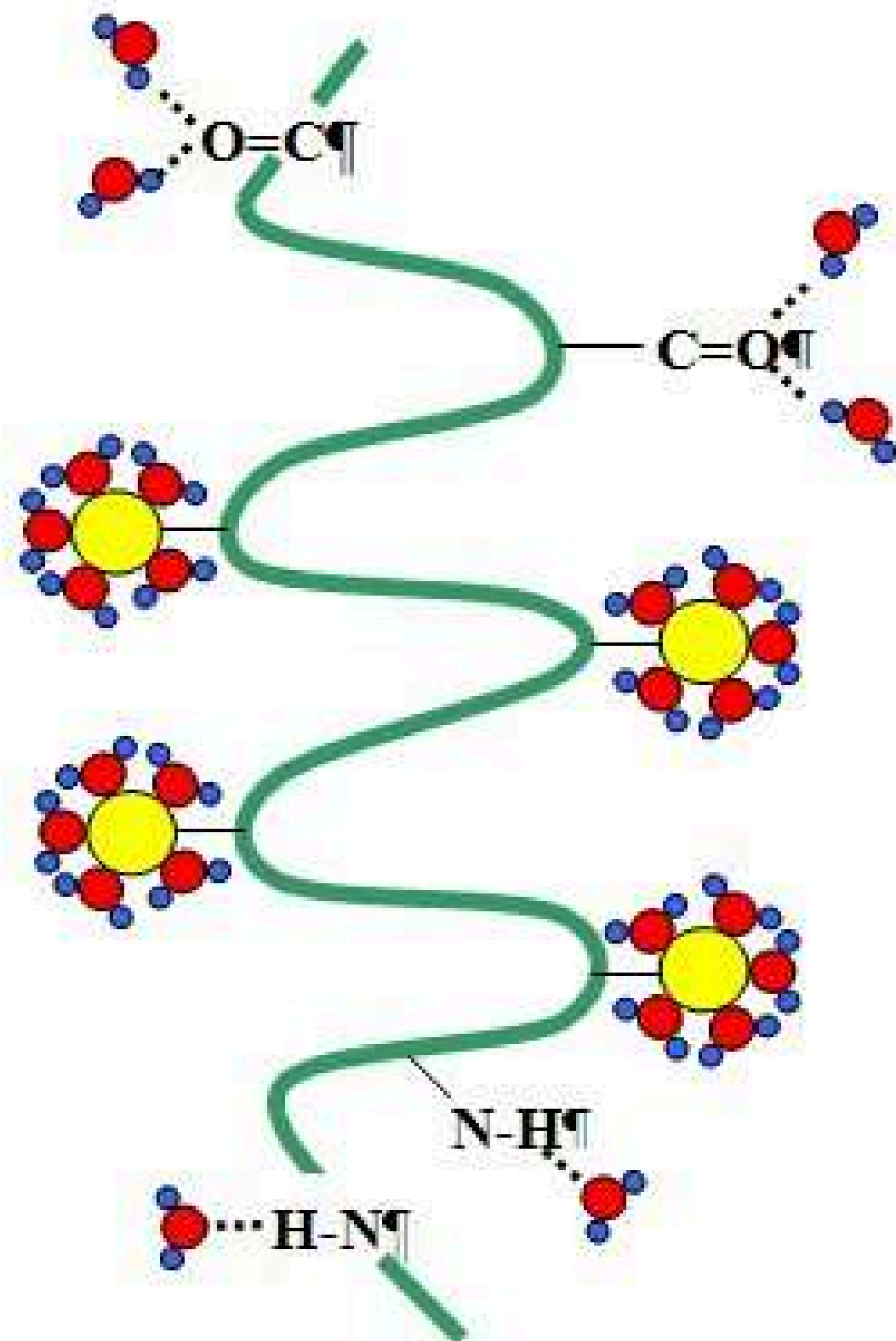
- **Antibody** purification is a critical step in obtaining highly pure antibodies for various applications in research, diagnostics, and therapy.
 - The process involves isolating and separating **antibodies** from complex mixtures, such as blood, serum, plasma, and cell culture supernatants, to obtain a highly enriched and purified antibody preparation.
-

AMMONIUM SULFATE PRECIPITATION

- The Ammonium sulfate precipitation method is a widely used method for antibody purification that exploits differences in protein solubility at different concentrations of ammonium sulfate.
 - Ammonium sulfate is a common salt used in protein purification due to its ability to selectively precipitate proteins from solutions based on their solubility characteristics.
-

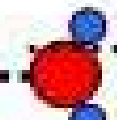
THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

- The solubility of proteins is influenced by their structure, surface charge, and the surrounding environment.
 - At low concentrations of **ammonium sulfate**, proteins remain soluble in solution due to the high water content and the electrostatic interactions between water molecules and charged residues on the protein surface.
-



 Hydrophobic group

 Hydrogen bond

 Water molecule

THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

- However, as the concentration of **ammonium sulfate** increases, it competes with water molecules for interaction with the protein surface.
 - This leads to the formation of salt bridges between **ammonium sulfate ions** and charged residues on the protein surface, reducing the protein's interaction with water molecules.
 - Consequently, the protein's solubility decreases, and it tends to aggregate and precipitate out of the solution.
-

THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

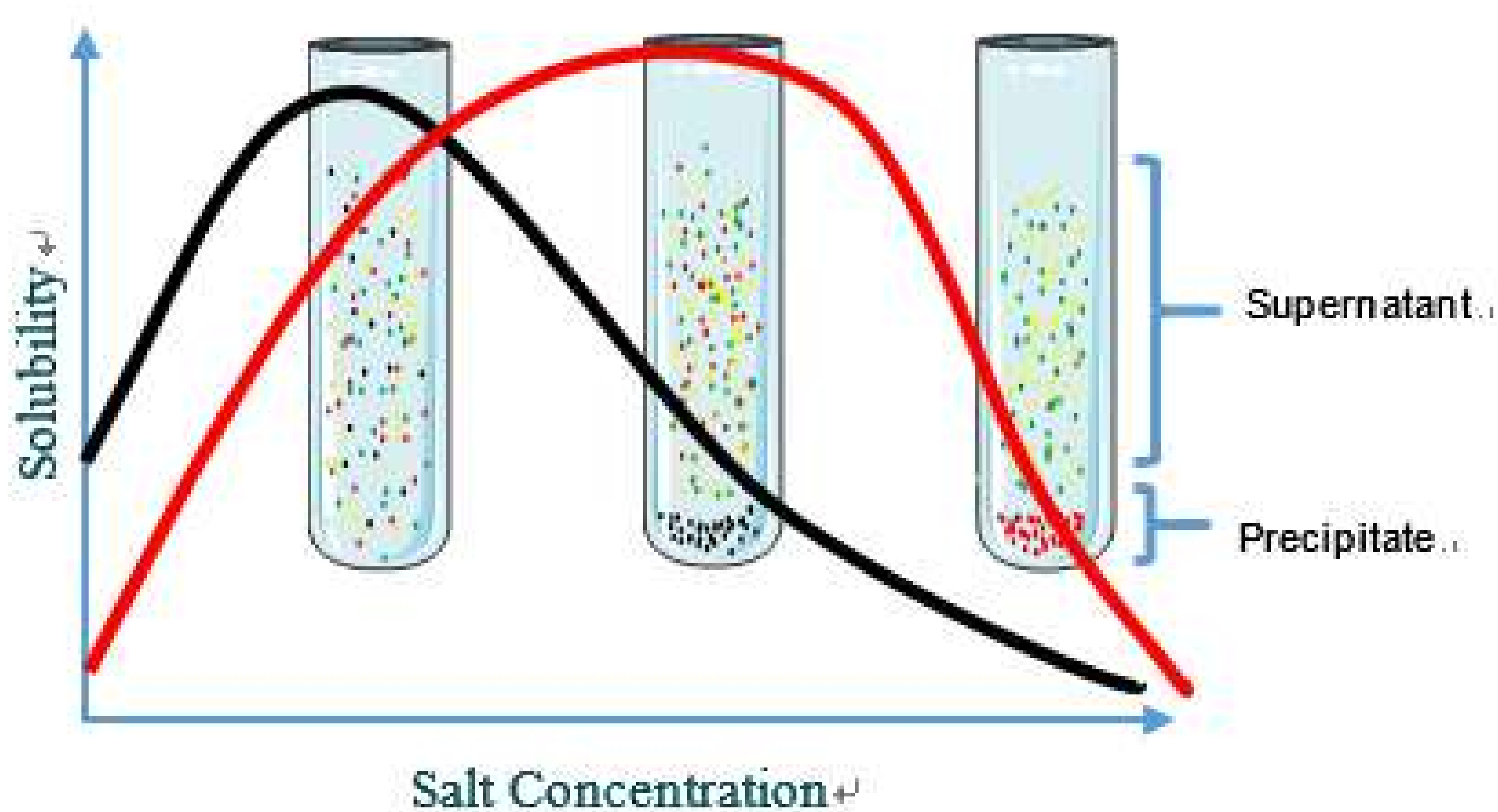
- The solubility of different proteins varies depending on their structural properties and amino acid composition.
 - Some proteins are more resistant to precipitation and require higher concentrations of **ammonium sulfate** to precipitate, while others precipitate at lower concentrations.
 - **Antibodies**, being globular proteins with specific structural features, often precipitate at intermediate concentrations of **ammonium sulfate**.
-

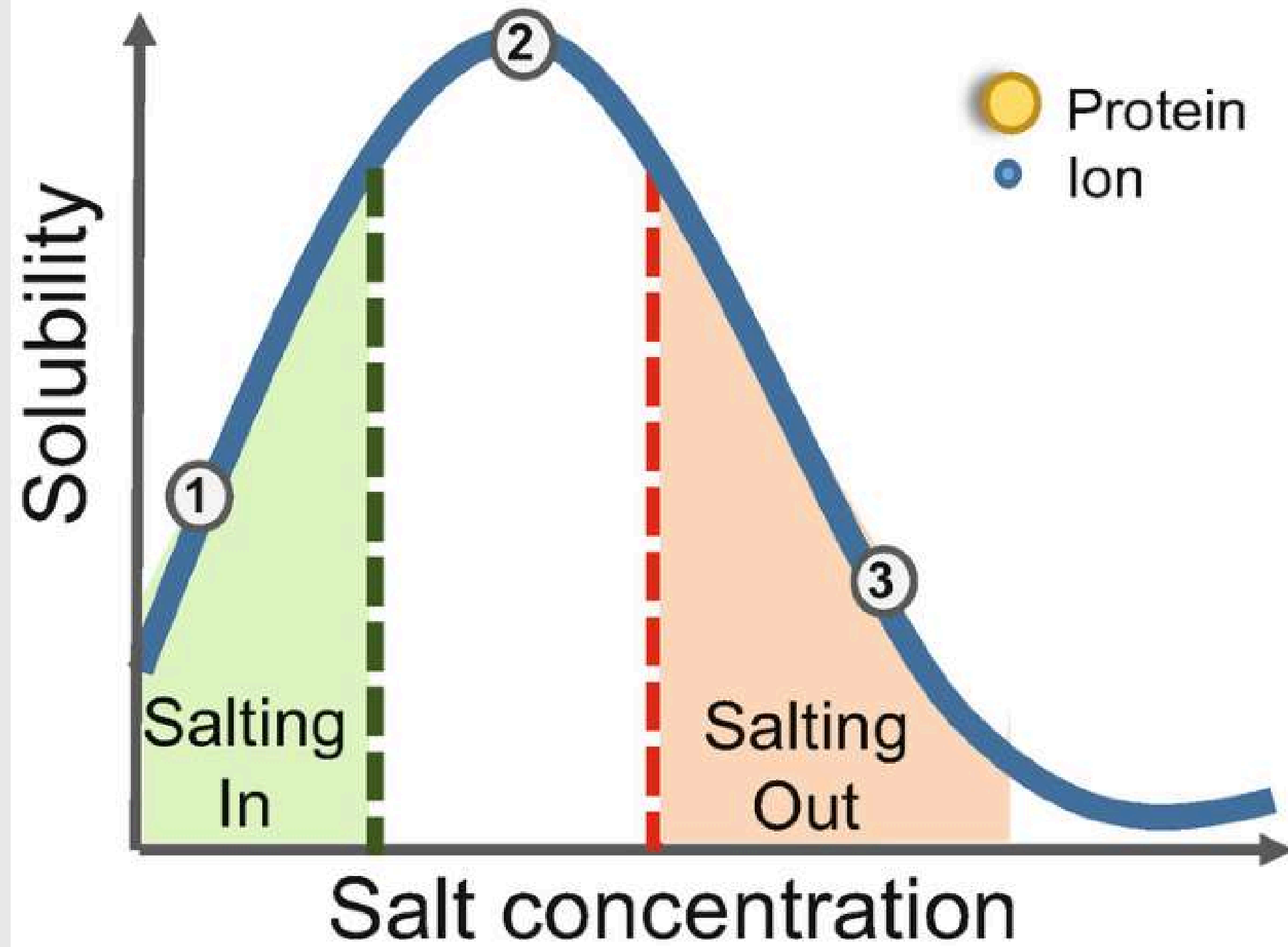
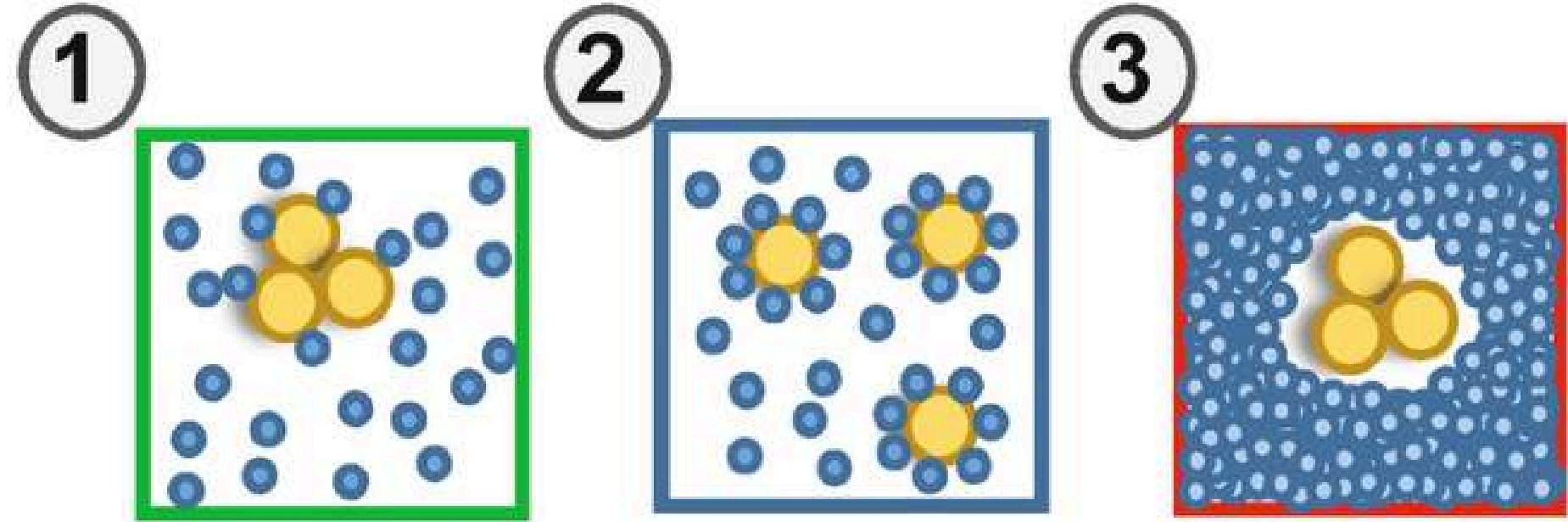
THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

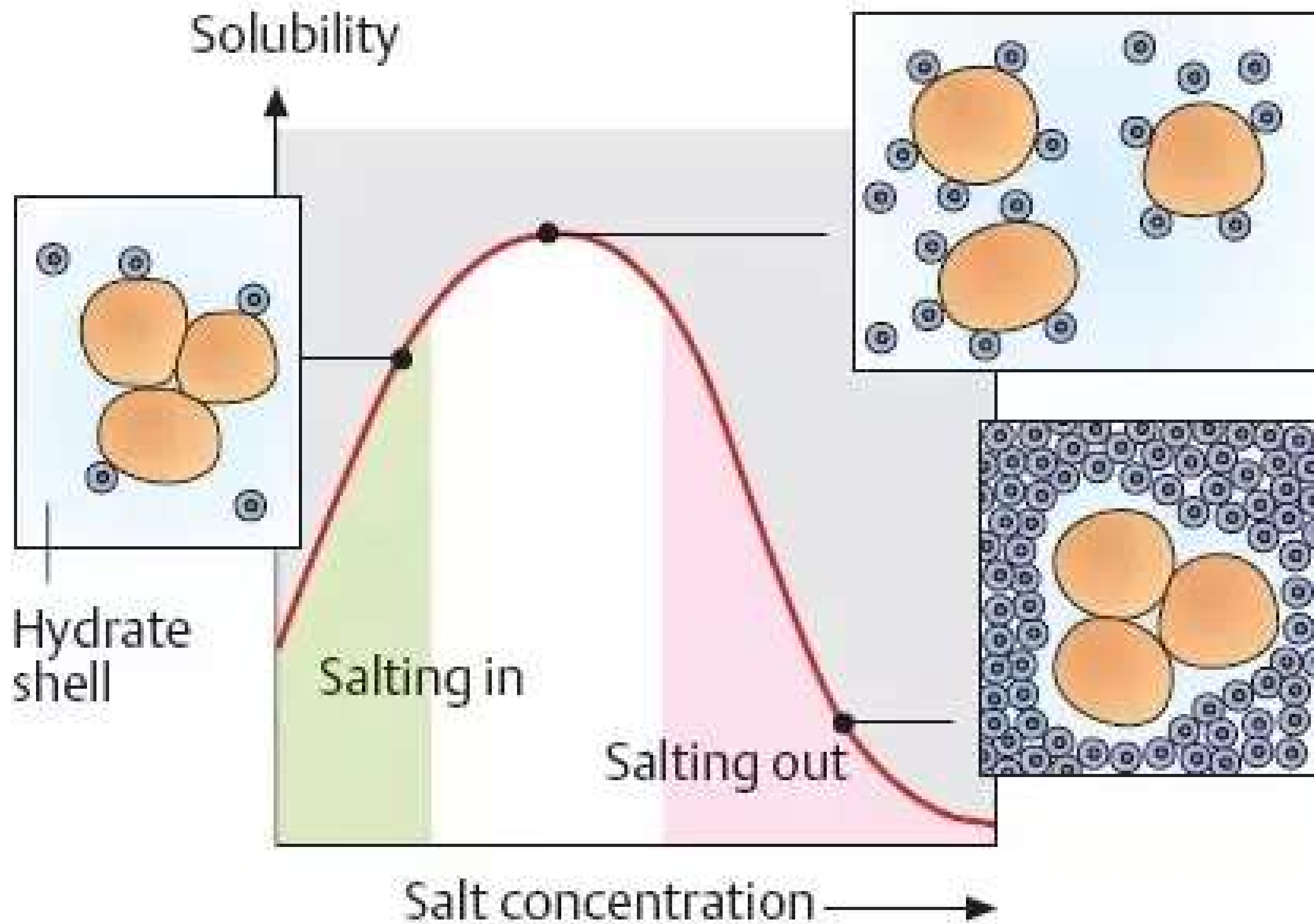
- By adjusting the concentration of **ammonium sulfate**, it is possible to selectively precipitate **antibodies** while leaving other proteins and contaminants in the solution.
 - After precipitation, the **antibody**-containing precipitate is separated from the soluble components by centrifugation, followed by resuspension and further purification steps to obtain highly purified **antibody** preparations.
-

THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

- By adjusting the concentration of **ammonium sulfate**, it is possible to selectively precipitate **antibodies** while leaving other proteins and contaminants in the solution.
 - After precipitation, the **antibody**-containing precipitate is separated from the soluble components by centrifugation, followed by resuspension and further purification steps to obtain highly purified **antibody** preparations.
-







THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

SALTING IN

- At low salt concentrations, ions in the solution can interact with charged groups on the surface of the antibody molecule, stabilizing its structure and enhancing its solubility.
 - This is because ions shield the electrostatic repulsion between charged residues on the protein surface, preventing protein-protein interactions and aggregation.
 - Thus, low concentrations of salt can "salt in" the antibody, increasing its solubility and stability in solution.
-

THE PRINCIPLE BEHIND ANTIBODY PRECIPITATION BY AMMONIUM SULFATE

SALTING OUT

- At high salt concentrations, ions in the solution can compete with water molecules for interaction with the protein surface.
 - As salt concentration increases, water molecules are displaced from the protein surface, leading to a decrease in the solvation and stability of the protein.
 - This disruption of the hydration shell around the protein can expose hydrophobic regions, causing protein-protein interactions and aggregation.
 - Ultimately, high salt concentrations can "salt out" the antibody, reducing its solubility and causing it to precipitate out of solution.
-

ADVANTAGES OF PRECIPITATION BY AMMONIUM SULFATE

- Ammonium sulfate precipitation is a simple and cost-effective method for initial antibody purification.
 - It can be used to quickly concentrate and partially purify antibodies from complex mixtures without the need for specialized equipment.
 - This method is compatible with a wide range of antibody sources, including blood, serum, plasma, and cell culture supernatants.
-

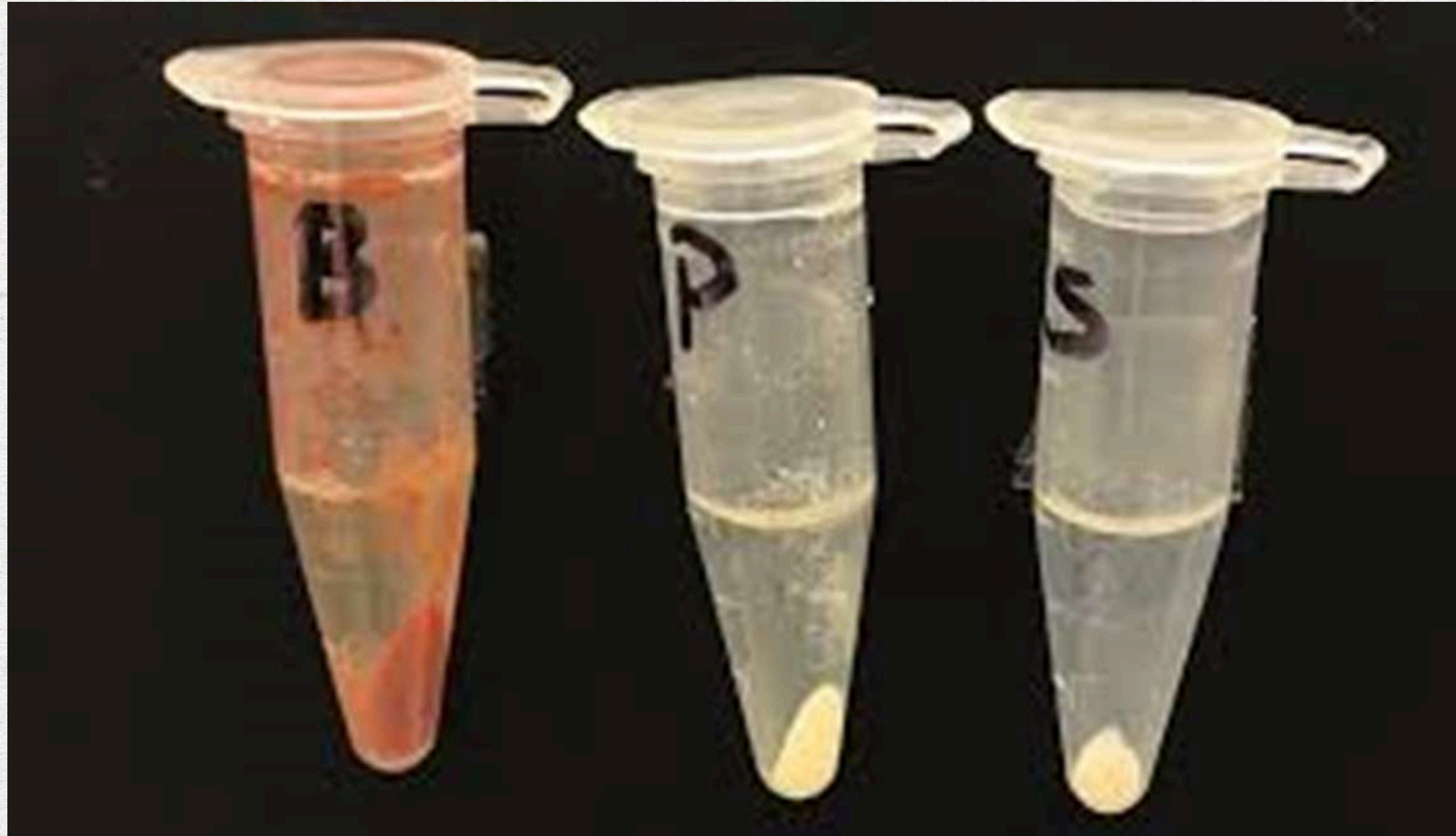
LIMITATIONS OF PRECIPITATION BY AMMONIUM SULFATE

- Ammonium sulfate precipitation may not provide high purity compared to more advanced chromatographic techniques.
 - It may result in loss of some antibody activity or specificity due to nonspecific binding or aggregation.
 - Optimization of the salt concentration is crucial to ensure selective precipitation of antibodies while minimizing precipitation of other proteins and contaminants.
-

PROCEDURE

1. Add separately, 1 ml of blood, plasma, and serum into a 15 ml conical tube.
 2. Gradually add 1m of Ammonium sulfate to the antibody-containing sample while stirring gently. It's crucial to add the ammonium sulfate slowly to avoid sudden changes in solution conditions.
 3. Once the total volume of ammonium sulfate is added, move the tube to 4°C for 6 hours or overnight to allow antibodies to precipitate.
 4. Transfer to the conical tube and centrifuge the precipitate at 5000 rpm for 30 minutes.
 5. Carefully remove and discard the supernatant. Invert the conical tube and drain well. For serum, resuspend the pellet in 30%–50% of the starting volume of 1X phosphate-buffered saline (PBS).
 6. store at -80°C for long-term storage.
-

PROTEIN PRECIPITATES AFTER CENTRIFUGE



CANVA LINK:

[https://www.canva.com/design/DAGARLzoKSw/GXnwN36GhOoZKFJXa5DJ4g/edit?
utm_content=DAGARLzoKSw&utm_campaign=designshare&
utm_medium=link2&utm_source=sharebutton](https://www.canva.com/design/DAGARLzoKSw/GXnwN36GhOoZKFJXa5DJ4g/edit?utm_content=DAGARLzoKSw&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton)

THE



END
